

Integrating Viterbi Dynamic Programming into SAVI

Canonical Prompting, Backpointers, and Conditional Trellis Inference

Revised Technical Design Memo — Companion to the SAVI Draft

Abstract

The current SAVI draft correctly frames semantic-state search as Viterbi-inspired, because ordinary LLM prompting conditions on the full text history and therefore violates the Markov property required by classical Viterbi inference. This memo integrates the useful parts of the Viterbi upgrade while avoiding an overclaim. The central change is *canonical prompting*: after merging histories that reach the same semantic state, the next LLM query is reconstructed from a deterministic representation of that state rather than from an arbitrary preserved text history. Under this induced process, SAVI can be implemented as a conditional Viterbi dynamic program over a beam-limited semantic-state trellis, with explicit backpointers and path recovery. The result is not unqualified exact Viterbi over the original LLM text process. It is exact only under stated conditions: sound semantic mapping, complete candidate support or accepted candidate truncation, no beam-pruning error, sufficient state expressiveness, and a calibrated learned value function for the bidirectional extension. The revised positioning is therefore stronger and safer: SAVI is Viterbi-inspired in its general form; *Canonical* SAVI implements conditional Viterbi DP over the induced semantic-state trellis.

Executive summary. Do not replace every occurrence of “Viterbi-inspired” with “exact Viterbi.” Instead, introduce **Canonical** SAVI as the principled variant. Canonical prompting makes the engineered inference process Markovian; DP backpointers make path recovery clean; and the correctness theorem should be conditional, not absolute.

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1 What Should Change in the SAVI Paper

1.1 The Safe Claim

The upgraded claim should be:

SAVI is Viterbi-inspired in its general form. Under canonical prompting, it becomes a conditional Viterbi dynamic program over an induced semantic-state trellis, exact up to candidate truncation, beam pruning, semantic-map soundness, state-representation expressiveness, and learned value approximation.

This is strong enough to improve the theory and safe enough to survive review.

1.2 Three Variants to Define

The paper should explicitly distinguish three variants.

Variant	Prompt source	Claim
History SAVI	Preserved representative text history.	Practical version. Viterbi-inspired, but Markov approximation may be nonzero.
Canonical SAVI	Deterministic prompt built from the canonical semantic state.	Conditional Viterbi DP over the induced semantic-state process.
Bidirectional Canonical SAVI	Canonical state prompt plus learned goal-conditioned value.	Forward Viterbi DP plus a learned approximation to the backward variable.

This resolves the main risk in the earlier memo: it preserves the Viterbi upgrade without claiming that the original history-conditioned LLM process is exactly Markovian.

2 Why the Original SAVI Is Only Viterbi-Inspired

Classical Viterbi inference requires a state abstraction that is sufficient for future transitions. More concretely, it requires:

- (i) **A state space.** There must be states $S_t \in \mathcal{S}$.
- (ii) **A transition process.** Future states must be induced by transitions from the current state.
- (iii) **A Markov property.** The distribution over future transitions must depend on the current state, not on the full history.

The original SAVI draft already has the first two ingredients: the semantic map Φ constructs states and candidate chunks induce transitions. The critical gap is the Markov property. LLMs condition on text histories. If two text histories H_A and H_B satisfy $\Phi(H_A) = \Phi(H_B)$, the LLM may still generate differently from them because the visible prompts differ.

Definition 1 (History-conditioned Markov error). For merged histories H_A, H_B with $\Phi(H_A) = \Phi(H_B)$, define

$$\varepsilon(H_A, H_B) = D[P_{\text{fwd}}(\cdot \mid H_A) \parallel P_{\text{fwd}}(\cdot \mid H_B)],$$

where D may be KL, JS divergence, or an empirical divergence over induced next semantic states.

When ε is large, merging is lossy: the algorithm discards one history even though that history may have produced better future continuations. This is why the base method should remain described as Viterbi-inspired.

3 Canonical Prompting: The Key Integration

Definition 2 (Canonical prompting). A prompting strategy is canonical if the LLM input at step t is a deterministic function of the current semantic state and the fixed problem statement:

$$\text{prompt}_t = f(S_{t-1}, \mathcal{P}),$$

independent of the text history used to reach S_{t-1} .

Under canonical prompting, two histories that merge into the same state produce the same next prompt. Thus, for the induced inference process,

$$P_{\text{fwd}}(x_t \mid H_A) = P_{\text{fwd}}(x_t \mid H_B) \quad \text{whenever} \quad \Phi(H_A) = \Phi(H_B),$$

provided both histories are replaced by the same canonical prompt. The Markov violation is not solved for the original text process; rather, the algorithm deliberately constructs a new state-conditioned process.

3.1 Canonical Prompt Templates

- **Symbolic algebra.** The prompt contains the original problem, the current canonical SymPy expression or equation, declared assumptions, and a fixed instruction to propose the next legal algebraic transformation. Prior derivation text is omitted.
- **Restricted code generation.** The prompt contains the function signature, problem specification, bounded observational state, visible output traces on fixed public inputs, and a fixed instruction to propose the next code chunk. Prior arbitrary prose is omitted or summarized into the canonical execution state.
- **Theorem proving.** The prompt contains the formal proof state, local hypotheses, goals, and allowed tactics. Prior natural-language proof text is not used as state.

Tradeoff. Canonical prompting removes path dependence, but it can lose useful context if the semantic state representation is too poor. This is not a theoretical footnote; it must be tested. The paper should add an experiment comparing canonical-state prompting against preserved-history prompting.

4 Viterbi DP with Backpointers

4.1 Forward Recurrence

Let $V_t(S)$ be the best accumulated log score of any path reaching canonical state S at step t . With canonical prompting, define

$$V_0(S_0) = 0, \quad V_0(S) = -\infty \text{ for } S \neq S_0.$$

The recurrence is

$$V_t(S') = \max_{\substack{S \in B_{t-1} \\ x: \Phi(S, x) = S'}} [V_{t-1}(S) + \log P_{\text{fwd}}(x \mid f(S, \mathcal{P}))]. \quad (1)$$

The backpointer is

$$\pi_t(S') = \arg \max_{(S, x): \Phi(S, x) = S'} [V_{t-1}(S) + \log P_{\text{fwd}}(x \mid f(S, \mathcal{P}))].$$

4.2 Bidirectional Extension

For Bidirectional Canonical SAVI, add a learned state value:

$$V_t^\lambda(S') = \max_{S, x: \Phi(S, x) = S'} [V_{t-1}^\lambda(S) + \log P_{\text{fwd}}(x \mid f(S, \mathcal{P})) + \lambda \log P_{\text{bwd}}(G = 1 \mid S', \mathcal{P})]. \quad (2)$$

This is not the exact HMM backward variable unless P_{bwd} is perfectly calibrated to the induced transition process. The safe statement is that P_{bwd} is a *learned approximation* to a backward value.

Definition 3 (Backward-value approximation). $P_{\text{bwd}}(G = 1 \mid S_t, \mathcal{P})$ estimates the probability that state S_t can reach a correct terminal solution under the induced decoding policy and task objective. It is a learned value function, not a future-token predictor.

4.3 Algorithm

Algorithm 1 Canonical SAVI as Conditional Viterbi DP

Require: Problem $\mathcal{P}$, semantic map Φ , LLM P_{fwd} , optional value P_{bwd} , candidates N , beam width K , steps T , weight λ

- 1: Initialize $V_0(S_0) \leftarrow 0$, $B_0 \leftarrow \{S_0\}$
- 2: **for** $t = 1, \dots, T$ **do**
- 3: $\mathcal{C} \leftarrow \emptyset$
- 4: **for each** $S \in B_{t-1}$ **do**
- 5: $p \leftarrow f(S, \mathcal{P})$ $\triangleright$ canonical prompt
- 6: Generate N candidate chunks $x_1, \dots, x_N \sim P_{\text{fwd}}(\cdot \mid p)$
- 7: **for each candidate** x_i **do**
- 8: $S' \leftarrow \Phi(S, x_i)$
- 9: **if** $S' \neq \perp$ **then**
- 10: $s \leftarrow V_{t-1}(S) + \log P_{\text{fwd}}(x_i \mid p) + \lambda \log P_{\text{bwd}}(G = 1 \mid S', \mathcal{P})$
- 11: Add (S', s, S, x_i) to $\mathcal{C}$
- 12: **end if**
- 13: **end for**
- 14: **end for**
- 15: **for each unique state** S' **in** $\mathcal{C}$ **do**
- 16: $V_t(S') \leftarrow \max\{s : (S', s, S, x) \in \mathcal{C}\}$
- 17: $\pi_t(S') \leftarrow \arg \max\{s : (S', s, S, x) \in \mathcal{C}\}$
- 18: **end for**
- 19: $B_t \leftarrow \text{top-}K \text{ states by } V_t(\cdot)$
- 20: **end for**
- 21: $S^* \leftarrow \arg \max_{S \in B_T} V_T(S)$
- 22: **return** trajectory recovered by backtracking through $\pi_T, \dots, \pi_1$

The key implementation changes are modest: prompt from canonical state, store backpointers, and recover the final trajectory by backtracking rather than by returning the last retained history.

5 Conditional Correctness Theorem

Theorem 1 (Conditional Viterbi correctness). *Assume the following conditions for the induced canonical-prompting process:*

- (i) **Canonical prompting:** all prompts are deterministic functions $f(S_t, \mathcal{P})$.
- (ii) **Sound semantic map:** Φ has no false merges over the evaluated domain.
- (iii) **Complete candidate support:** the candidate generator enumerates every transition with nonzero probability, or else the theorem is restricted to the sampled candidate set.
- (iv) **No beam-pruning error:** the optimal path's state remains within the top- K retained states at every step.

- (v) **For bidirectional SAVI only:** P_{bwd} is a calibrated estimate of terminal success probability under the induced process.

Then Algorithm 1 with $\lambda = 0$ recovers the maximum-score path in the induced semantic-state trellis. With $\lambda > 0$, it recovers the maximum-score path under the augmented forward-plus-value objective in Equation 2.

Proof sketch. Under canonical prompting, future candidate distributions depend only on the current canonical state. Under semantic-map soundness, merging preserves equivalence classes rather than collapsing distinct states. Equation 1 is therefore the standard Viterbi max recurrence over the induced trellis. Backpointers recover the maximizing path. Candidate truncation and beam pruning restrict the trellis on which the theorem applies; if the optimal path is pruned or never sampled, no beam method can recover it. The bidirectional case is exact only for the stated augmented objective and depends on the calibration quality of P_{bwd} . $\square$

Remark 1. *This theorem is intentionally conditional. That is a feature, not a weakness. Each condition corresponds to an empirical audit: false-merge rate, candidate budget, beam-width sensitivity, canonical-prompting validation, and value calibration.*

6 How to Modify the Paper Text

6.1 Abstract and Contributions

Replace any unqualified claim of “exact Viterbi” with this language:

SAVI is Viterbi-inspired in its general form. We further introduce Canonical SAVI, which reconstructs the LLM prompt from the canonical semantic state after each merge. This induces a state-conditioned process on which SAVI can be implemented as a conditional Viterbi dynamic program with explicit backpointers and path recovery.

Add a contribution bullet:

We define Canonical SAVI and prove a conditional correctness theorem showing that, under sound semantic mapping, sufficient candidate support, and no beam-pruning error, the algorithm recovers the maximum-score path in the induced semantic-state trellis.

6.2 Framework Section

Add a new subsection after the semantic map definitions:

Canonical prompting. The base form of SAVI may continue from the representative text history associated with each retained state. This is practical but only approximately Markovian. Canonical SAVI instead constructs the next prompt from the canonical state itself. This enforces the beam invariant not only at the state level but also at the prompting level: all paths that merge into the same state produce the same future prompt.

Then replace the recurrence with Equation 1 and introduce the backpointer table π_t .

6.3 Theory Section

Do not delete the Markov approximation discussion. Split it:

1. **History SAVI:** Markov approximation error must be measured because prompts differ across merged histories.
2. **Canonical SAVI:** Markov error is removed by construction for the induced process, but the burden shifts to state expressiveness and semantic-map soundness.

This is better than claiming the Markov problem vanishes universally.

6.4 Bidirectional SAVI Section

Replace “ P_{bwd} is the true backward variable” with:

P_{bwd} is a learned approximation to the backward value: the probability that a canonical state can reach a correct terminal solution under the induced decoding process. If calibrated, it plays the role of the HMM backward variable; otherwise it should be treated as a value heuristic.

7 Experiments to Add

The Viterbi integration needs experiments that specifically test whether the theory helps, not just whether the original semantic merging idea works.

Experiment	Comparison	Purpose
E6a Canonical prompt validation	History SAVI vs Canonical SAVI.	Tests whether state-only prompting loses useful information. Success: canonical accuracy within 2–3 percentage points, or better after enriching the state representation.
E6b Markov reduction	Next-state distributions from two merged histories before and after canonical prompting.	Shows that canonical prompting collapses history-dependent divergence in the induced process.
E6c DP backtracking ablation	Beam-with-merge vs Viterbi DP with backpointers.	Tests whether principled path recovery improves performance, especially on longer chains.
E6d State expressiveness ablation	Minimal state prompt vs enriched state prompt.	Identifies whether canonical prompting fails because the state representation omits necessary context.
E6e Value calibration	Reliability curve and expected calibration error for P_{bwd} .	Determines whether Bidirectional SAVI can legitimately be interpreted as using a backward-value approximation.

These should be added to the experimental section after the existing aliasing and online-vs-post-hoc experiments. They do not replace BAR, CMR, RSD, false-merge audits, or iso-compute comparisons.

8 Revised Nomenclature

Avoid	Use instead
“SAVI is exact Viterbi”	“Canonical SAVI is conditional Viterbi DP over the induced semantic-state trellis.”
“Remove Viterbi-inspired everywhere”	“Base SAVI is Viterbi-inspired; Canonical SAVI is the Viterbi-DP variant.”
“ P_{bwd} is the true backward variable”	“ P_{bwd} is a learned approximation to the backward value, calibrated against terminal success.”
“Markov error collapses to zero”	“Canonical prompting removes history dependence in the induced process; residual risk is state expressiveness and Φ soundness.”
“Viterbi optimality”	“Optimality within the sampled, beam-limited, soundly merged semantic trellis.”

9 Risk Analysis

Risk	Problem	Mitigation
Canonical prompting loses context	The semantic state may omit information contained in the original text history.	Run E6a and E6d. Enrich the state prompt with assumptions, variable bindings, constraints, and selected verified provenance if needed.
False merges corrupt DP	Viterbi DP assumes that merged paths are genuinely equivalent.	Keep existing false-merge audit; report false-merge and missed-merge rates separately for algebra and code.
Candidate truncation weakens exactness	Sampling only N chunks per state means the full transition distribution is not enumerated.	State that the theorem applies to the sampled trellis; sweep N to measure sensitivity.
Beam pruning drops optimal states	Top- K pruning can eliminate the globally best path.	Sweep K and report performance/coverage curves.
P_{bwd} is uncalibrated	An uncalibrated value model can harm Bidirectional SAVI.	Add E6e; report calibration and perform lambda sensitivity. Keep Forward SAVI as the main contribution.

10 Recommended Integration Order

1. Add Canonical SAVI as a named variant, not as a replacement for all SAVI.
2. Modify the algorithm to include canonical prompting, DP table updates, and backpointers.
3. Add the conditional correctness theorem with explicit assumptions.
4. Add the experiments E6a–E6e to validate canonical prompting and DP backtracking.

5. Soften the P_{bwd} language to “learned approximation to the backward value.”
6. Keep Forward SAVI as the main empirical contribution until P_{bwd} is validated.

Final recommended sentence for the paper. SAVI converts text-space tree search into semantic-state trellis search. In its practical history-conditioned form it is Viterbi-inspired; in its canonical-prompting form it implements conditional Viterbi DP over the induced semantic-state trellis, with approximation sources made explicit and empirically measurable.